

CMSE 381 Study Guide

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Here we summarize the detailed learning goals in terms of a series of questions that you should be able to answer at the end of this course.

(Work in progress... Last update 3/18/2026)

Part 1

Intro to statistical learning

- Regression vs classification
- Input vs output variable
- Setup for model $Y = f(X) + \varepsilon$
- Reduceable vs irreducible error
- Training vs testing data
- Parametric vs non-parametric
- Flexibility vs interpretability
- Supervised vs unsupervised learning
- Quantitative vs Qualitative variable
- Mean squared error (training vs testing)
- Bias variance tradeoff

Linear Regression

- Setup
- Least squares coefficient estimates
- Standard error for β_i
- CI for β_i
- Hypothesis test
- Residual standard error
- R-squared
- Multiple linear regression
- Interaction terms
- Dummy variables

Classification (Some from ch 2)

- Error rate for classification
 - Bayes Classifier
 - Bayes error
 - KNN
 - Logistic Regression
 - Log odds (logits)
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Part 2

Methods of validations

(Required readings: chapter sections 5.1 ... 10 pages)

- Why do you need validation methods?
- What is the difference between validation scores and true test errors?
- What are the three most basic validation methods?
- For each of these methods,

- How do you calculate the validation score? You should be able to describe the procedures verbally and mathematically.
- How do you calculate the validation score in Python?
- What are the advantages and disadvantages of using this method compared to the two other methods? You should be able to describe them in terms of computational cost and bias/variance of estimation.
- How do you use validation methods to select the appropriate meta-parameter?
 - Given a plot of validation score as a function of model flexibility, which depends on a specific meta-parameter of the model, you should be able to choose the right parameter by reading the plot, and provide an explanation based on the principle of bias-variance tradeoff.
 - Given a dataset, you should be able to generate this plot yourself in Python for any type of model covered in this class.

Subset/feature selection

(Required readings: chapter section 6.1 ... 9 pages)

- Why shouldn't you have as many predictors in your model as possible?
- What are the three basic methods that could be used to select the appropriate predictors (feature selection)?
- What are the steps/procedures in the algorithm for each of these methods?
- How to implement these procedures by hand given the training and CV test error for each combo of predictors?
- What are the pros and cons for choosing each of these feature selection methods? When can you or cannot use them?
- How do you implement the feature selection algorithm in Python?

Regularization methods

(Required readings: chapter section 6.2.1-6.2.2 ... 12 pages)

- What is regularization? Why do we need it?
- What are the two basic types of regularization methods? How are they implemented mathematically in linear regression? Why are they also called Shrinkage methods?
- How do you fit these two basic types of regularized linear regression model in Python?

- How do you control the model flexibility & bias-variance tradeoff when using regularization?
- How do you find the right amount of regularization using cross-validation? How do you do this in python?
- What additional precautions do you need to take when using regularization (compared to least squares)?
- What are the advantages of regularization compared to Least Squares?
- What are the advantages of regularization compared to subset selection?
- What are the advantages of one type Shrinkage method over another? When do you choose one over another?

Dimensionality reduction and feature engineering

(Required readings: chapter section 6.3 ... 9 pages)

- How to create new variables as linear combinations of the original predictors?
- Why do we need **Principal Component Analysis (PCA)**? What is the main purpose of using it?
- What is a principal component (PC)?
 - What does the first PC maximize? You should be able to explain this both geometrically in a plot and mathematically.
 - Similarly, what do the subsequent PCs maximize?
- How do you compute the PCs in Python, given a dataset?
 - How do you project data points on each PC? You should also be able to plot the data points in the PC space.
 - How do you find out how much variance each PC explains?
- Why do you want to use the PCs in regression models?
 - What assumptions do you have to make for it to be a good idea to use **principal component regression (PCR)**?
 - Or conversely, what is a typical bad scenario to use PCR?
- How do you implement PCR in Python?
- How do you interpret the model coefficients when using PCR?
- How do you choose the number of PC to use in PCR?

- Given a figure of, e.g., cross-validation score as a function of the number of PCs, you should be able to choose appropriately and provide rationales in terms of bias-variance tradeoff.
 - You should be able to generate such figures in Python.
 - What is the relationship/differences between PCR and feature selection and regularization methods that you learned in this part of the course?
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Part 3

Polynomial functions and step functions

(Required readings: chapter section 7.1-7.2 ... 4 pages)

- Why do people want to use polynomial regression rather than simple linear regression?
- How to fit polynomial regression models?
 - You should be able to describe the procedures verbally and implement them in Python
- Why do people want to use step functions rather than simple linear regression?
- How to fit step function models?
 - You should be able to describe the procedures for both regression and classification problems.
 - You should also be able implement them in Python.
- What is common between polynomial regression and step functions?
- What is an advantage of using step functions than polynomial despite its simplicity?
 - You should be able to distinguish between models with equations that apply to the whole domain vs those that don't.

Basis Functions and Splines

(Required readings: chapter section 7.3-7.4 ... 7 pages)

- What are basis functions?
 - You should be able to articulate what are the basis functions b_i for different models that we have covered.

- Examples include polynomial, step functions, and cubic splines.
- What are the purpose of using basis functions?
 - How do they influence model flexibility?
 - How does this purpose manifest in different models, such as regression splines?
- How to define cubic splines mathematically in terms of piecewise polynomial?
 - You should be able to write down the equations for the model between knots
 - the equations for constraints at the knots
 - and the total degrees of freedom
- How to calculate the cubic spline coefficients by hand by applying the above mathematical definitions?
- How to define cubic splines in terms of basis functions?
 - You should be able to describe it mathematically in terms truncated power basis function (textbook version).
 - You should be able to visualize them as B-spline basis function in Python.
- How do you fit a cubic spline model to data in Python?
- How do you change the model flexibility?
 - What are the relevant metaparameters?
 - How do you choose the appropriate flexibility?
- What precautions must you take at the outer boundary? Why?

Decision Trees and Random Forests

(Required readings: chapter section 8.1-8.2.2 ... 16 pages)

- How does a decision tree make decisions?
 - Given a tree and a data point with multiple predictors $(x_1, x_2, \dots, x_p)$, you should be able to derive what the predicted outcome $\hat{y}$ should be.
 - You should be able to do this for both regression and classification decision trees.
- What are different parts of a decision tree called? What do they represent?
 - You should be able to point out what are: leaves (terminal nodes), internal nodes, branches/edges.
 - Given a simple example (e.g., two predictors), you should be able to point out which region in the predictor space map to which leave.

- How do you fit a decision tree?
 - What is the cost function for regression and classification tree respectively?
 - How does recursive binary splitting work?
 - How does it end?
 - How do you do it in Python? How do you visualize the tree in Python?
- What meta-parameter of decision trees determine their flexibility?
- Why do you need to prune the tree?
- What are the advantages and disadvantages of using decision trees vs. linear regression?
- What does bagging of decision trees accomplish?
- How do you use out of bag error estimation for decision trees?
 - You should be able to describe this for both regression and classification trees.
- What problem of bagging does using random forest address?
- What is the relationship between random forest and bagging?

Maximal Margin Classifiers and Support Vector Classifier/Machines

(Required readings: chapter section 9.1-9.4 ... 17 pages)

- What is hyperplane?
 - How do you mathematically describe this hyperplane?
 - How do you mathematically describe the two sides of the hyperplane?
 - Given the equation of the hyperplane and the coordinates for a point, you should be able to tell which side of the plane the point is on.
- What qualify a hyperplane as a separating hyperplane?
 - You should be able to describe this mathematically using an inequality.
 - You should also be able to determine whether a hyperplane is a separating hyperplane given a graph, or given the equation of the plane and the coordinates and class of a few points.
- How to use a separating hyperplane as a classifier?
- What makes a hyperplane a maximal margin hyperplane?
 - What are its margin and support vectors?

- Given a graph, you should be able to clearly label the margin and support vectors. You should also be able to infer the size of the margin from reading the graph.
- You should also be able to describe the optimization problem mathematically.
- Given the equation of the maximal margin hyperplane and the coordinates of support vectors, you should be able to calculate the size of the margin by hand.
- What is the difference between Support Vector Classifier and Maximal Margin Classifier?
 - What short-comings of Maximal Margin Classifier does Support Vector Classifier overcome?
 - How?
 - You should be able to describe the difference verbally, graphically, and mathematically.
- How to interpret each element of the mathematical formulation of Support Vector Machine graphically?
 - Given a graph of Support Vector Classifier and some data points, you should be able to derive the values of different parameters in the mathematical formulation (e.g. M , ϵ_i).
- What parameters controls the flexibility (hence bias-variance trade-off) of Support Vector Classifier?
- What are the support vectors of Support Vector Classifier?
- How to use Support Vector Classifier in python?
 - You should be able to obtain and interpret the fitted parameters.
 - You should be able to select the appropriate hyperparameter that optimize bias-variance trade off.
- How is nonlinearity introduced in Support Vector Machine?
 - You should be able to describe how kernels work conceptually and mathematically.
- What type of kernel makes Support Vector Machine equivalent to Support Vector Classifier?
- What other kernels are there?
- What type of kernel leads to more local behavior?
- How to use Support Vector Machine in Python?
 - You should be able to fit a Support Vector Machine using different kernels covered in class.

- You should be able select the appropriate hyperparameter that optimize bias-variance trade off.

Neural Networks

(Required readings: chapter section 10.1-10.3 ... 14 pages)

- What is the architecture of 1-layer feed-forward neural network?
 - You should be able to describe what happens at each layer (e.g., input layer, hidden layer, output layer) conceptually and mathematically.
- What is activation? What types of activation functions are there?
 - You should be able to explain it mathematically.
 - Given example inputs and weights from the previous matrix, you should be able to calculate the activation A by hand (with the help of a calculator).
- How does a 1-layer feed-forward neural network produce an output?
 - Given a β matrix and activation from the previous layer, you should be able to calculate the output by hand.
- What do the fitted parameters minimize in a 1-layer feed-forward neural network?
- What is the architecture of multilayer feed-forward neural networks?
 - You should be able to describe what happens at each layer mathematically for a typical classification problem.
 - You should be able to calculate class probability for the output layer using softmax.
- How to classify images using multilayer neural networks in Python?
- What is the architecture of a convolutional neural network?
 - You should be able to describe the types of layers, how they are arranged, and the purpose of each of them.
- How does convolution work?
 - You should be able to compute the result of the convolution between a simple matrix and a simple filter by hand.
- What information does Convolutional Neural Network capture through convolution?

Unsupervised learning / Clustering

(Required readings: chapter section 12.1, 12.4 ... 16 pages)

- What is the difference between supervised vs unsupervised learning?
- What do clustering methods aim to accomplish?
- How to interpret a dendrogram of hierarchical clustering?
- How are different linkage methods defined?
- How to perform hierarchical clustering in Python?